

EC2 — 25 Practice Tasks (Beginner Junior)

Beginner-friendly, hands-on tasks. Each lists a **Goal**, **Do** (hints/commands), and **Done when** (how you know it's complete). Record commands/screenshots for submission.

Free Tier: use `t2.micro` / `t3.micro`. **Terminate instances and delete stray volumes/EIPs after each task.** Set a billing budget alarm first.

Legend: Beginner · Junior

Beginner (1–14)

Task 1 — Launch your first instance

Goal: Boot an Amazon Linux 2023 `t3.micro`. **Do:** Create a key pair, accept the default VPC, 8 GB gp3. **Done when:** Instance state is `running` and you saved the `.pem` safely.

Task 2 — Connect over SSH

Goal: Log in from your terminal. **Do:** `chmod 400 key.pem ; ssh -i key.pem ec2-user@PUBLIC_IP`. **Done when:** You get a shell prompt on the instance.

Task 3 — Identify the core parts

Goal: Understand what makes an instance. **Do:** Note the AMI, instance type, key pair, security group, and EBS volume in the console. **Done when:** You can explain what each one is.

Task 4 — Restrict SSH to your IP

Goal: Basic security. **Do:** Edit the security group so port 22 is allowed from **My IP** only. **Done when:** SSH works from your IP; you explain why `0.0.0.0/0` on 22 is risky.

Task 5 — Allow HTTP

Goal: Prepare to serve web traffic. **Do:** Add an inbound rule for port 80 from anywhere. **Done when:** The rule appears in the security group.

Task 6 — Serve a web page

Goal: Run a web server. **Do:** `sudo dnf install -y httpd && sudo systemctl enable --now httpd`; write an index page. **Done when:** `http://PUBLIC_IP/` shows your page.

Task 7 — Reboot the instance

Goal: Understand reboot. **Do:** Reboot from the console; reconnect. **Done when:** The instance comes back with the same public IP.

Task 8 — Stop and start

Goal: See lifecycle behavior. **Do:** Stop the instance, then start it again. **Done when:** You notice the public IP changed and can explain why.

Task 9 — Read status checks

Goal: Health basics. **Do:** Look at the 2/2 status checks in the console. **Done when:** You explain system vs instance checks.

Task 10 — Tag your resources

Goal: Organization. **Do:** Add `Name` , `Owner` , and `Environment=learning` tags. **Done when:** All tags appear on the instance.

Task 11 — View instance metadata

Goal: Self-awareness from inside. **Do:** `TOKEN=$(curl -X PUT "http://169.254.169.254/latest/api/token" -H "X-aws-ec2-metadata-token-ttl-seconds: 60")` ; then curl the instance-id with that token. **Done when:** You retrieve the instance-id and AZ.

Task 12 — Understand regions & AZs

Goal: Know where things live. **Do:** Note your instance's region and AZ; switch the console region and see it "disappear". **Done when:** You explain region-scoping.

Task 13 — Terminate an instance

Goal: Clean shutdown. **Do:** Terminate the instance from the console. **Done when:** State becomes `terminated` .

Task 14 — Cleanup check

Goal: Cost hygiene. **Do:** Confirm no stray volumes or Elastic IPs remain. **Done when:** The console shows nothing left running/allocated.

Junior (15–25)

Task 15 — User data bootstrap

Goal: Auto-configure at launch. **Do:** Launch a new instance with user data that installs httpd and writes a page. **Done when:** The page is live without you SSHing in.

Task 16 — Create and use a separate key pair

Goal: Key management. **Do:** Create a named key pair; launch an instance with it; connect. **Done when:** You log in with the new key.

Task 17 — Attach and mount an EBS volume

Goal: Add storage. **Do:** Create a 5 GB gp3 volume, attach it, `mkfs.ext4` , mount at `/data` . **Done when:** `df -h` shows `/data` .

Task 18 — Persistent mount

Goal: Survive reboot. **Do:** Add the volume to `/etc/fstab` with `nofail` ; `sudo mount -a` . **Done when:** `/data` remounts without error.

Task 19 — Create an EBS snapshot

Goal: Back up a volume. **Do:** Take a snapshot of the volume in the console. **Done when:** Snapshot status is `completed` .

Task 20 — Create a custom AMI

Goal: Reusable image. **Do:** Configure an instance (httpd + page), then create an AMI from it. **Done when:** A new instance from your AMI already has the app.

Task 21 — Elastic IP

Goal: Stable public IP. **Do:** Allocate and associate an Elastic IP; stop/start and confirm it stays. **Done when:** IP survives stop/start (then release it — idle EIPs cost money).

Task 22 — Resize the instance type

Goal: Change capacity. **Do:** Stop the instance, change type (e.g. `t3.micro` → `t3.small`), start. **Done when:** The instance runs on the new type.

Task 23 — List instances via CLI

Goal: CLI basics. **Do:** `aws ec2 describe-instances --query 'Reservations[].Instances[].[InstanceId,State.Name]' --output table` . **Done when:** Your instance appears in the output.

Task 24 — Start/stop via CLI

Goal: Automate simple actions. **Do:** `aws ec2 stop-instances` / `start-instances --instance-ids i-...` . **Done when:** The CLI changes the instance state.

Task 25 — View CPU in CloudWatch

Goal: Basic monitoring. **Do:** In CloudWatch Metrics EC2, graph `CPUUtilization` . **Done when:** You see the CPU graph for your instance.

Cleanup after each task

Terminate instances, delete unattached volumes, release Elastic IPs. Verify with `aws ec2 describe-instances` , `describe-volumes` , `describe-addresses` .

Submission

Per task: command(s)/console steps, a screenshot or CLI output proving "Done when", and one sentence on what you learned.
Grade against `.../EVALUATION-CHECKLIST.md` .